

Mersenne Exponents p - Geometric Square App

Mersenne Geometric Square

Corners = seeds {2,3,5,7} · Click = buttons to compute

2

TL

3

TR

center

7

BL

5

BR

Diagonals

TL × BR = 10

TR × BL = 21

Corner × N + Add

TL × 1 + 1 = 3

Generator

TR × BR = -

Gen × TL + 1 = -

Sum / - Diff

TL + TR = 5

2 + 3 = 5

TL × 1 + 1 = 3

History

Clear History

We build the seeds:

$TL + 1 = 3 = TR$
 $2 + 1 = 3$
 $TL + TR = 5 = BR$
 $2 + 3 = 5$
 $TL + BR = 7 = BL$
 $2 + 5 = 7$

Sum / - Diff

TL + BR = 7

Corner × N + Add

TL × 3 + 7 = 13

Corner × N + Add

TR × 5 + 2 = 17

$$TL \times TR + BL = 13$$

$$| \quad TR \times BR + TL = 17$$

We have also diagonal $TL \times BR = 10$

$$10 + TR = 13, (10+3 = 13)$$

$$10 + BL = 17, (10+7 = 17)$$

We have also $TL + TR + BR + BL = 17$


Corner × N +
Add

BR ▾ × 3 + 2

=

17


Generator

TL ▾ × TR ▾ = -

Gen × TL ▾ + 1

= 13


Corner × N +
Add

BL ▾ × 2 + 3

=

17


Generator

TL ▾ × TR ▾ = -

Gen × TR ▾ + 1

= 19

$$10 \text{ (diagonal)} \times 3 + 1 = 31 \text{ or } TL \times TR \times BR + 1 = 31, (2 \times 3 \times 5 + 1 = 31)$$


Generator

TL ▾ × TR ▾ = -

Gen × BR ▾ + 1

= 31


Corner × N +
Add

TR ▾ × 10 + 1

=

31


Corner × N +
Add

TR ▾ × 10 + 31

=

61

$$TL \times 21 \text{ (diagonal)} + 19 = 61$$


Diagonals

TL × BR = 10

TR × BL = 21


Corner × N +
Add

TL ▾ × 21 + 19

=

61

▲ Generator

BL ▾ × BR ▾ = -

Gen × TL ▾ + 19

= 89

▲ Corner × N + Add

BR ▾ × 21 + 2

= 107

BL × BR × TL + 19 = 89
 (7×5×2 + 19 = 89)

| BR × 21 (diagonal) + 2 = 107

▲ Generator

BL ▾ × BR ▾ = -

Gen × TR ▾ + 2

= 107

▲ Generator

BR ▾ × BR ▾ = -

Gen × BR ▾ + 2

= 127

BL × BR × TR + 2 = 107
 7×5×3 + 2 = 107

| BR × BR × BR + 2 = 127
 | 5×5×5 + 2 = 127

Mersenne Geometric Square

Corners = seeds {2,3,5,7} · Click = buttons to compute

2 TL 3 TR

center

13 BL 7 BR

Diagonals

TL × BR = 14

TR × BL = 39

Corner × N + Add

BL ▾ × 31 + 127

= -

Generator

TR ▾ × BL ▾ = -

Gen × BL ▾ + 14

= 521

+ Sum / - Diff

TL ▾ + ▾ BR ▾

= -

39 × BL + 14 = 521

We rotate the p number in the square. TR × BL × BL + 14 (diagonal) = 521

